VRANDESH BANDIKATTI

FULL STACK DEVELOPER | TYPESCRIPT / NODE.JS / REACT | AWS / AZURE / GCP | AI-NATIVE PLATFORMS

Melbourne, Australia | 0468 450 792 | vrandesh@gmail.com | linkedin.com/in/vrandesh

PROFILE

Senior Full Stack Tech Lead and Engineering leader with 16+ years of software engineering experience designing and delivering production SaaS platforms across enterprise data, payments, workflow automation and cloud-native applications. Strong hands-on background in Python, FastAPI, Django, REST/GraphQL APIs, distributed services, event-driven architectures and workflow orchestration.

Experienced building production systems incorporating asynchronous processing, stateful workflows, retries, idempotency, rate limiting, caching, authentication, observability and automated CI/CD. Delivered microservice and serverless architectures across GCP, AWS and Azure, including Cloud Functions/Firebase, AWS Lambda/Step Functions and Azure Durable Functions.

Built and maintained 12+ production microservices and complex enterprise integrations across financial, data and analytics platforms. Recent work includes Python-based orchestration pipelines, FastAPI services, high-volume document-processing workflows, scheduled data processing, governed data integrations and automated production workloads.

Strong understanding of cloud-native engineering, Docker-based deployment, API integration, data pipelines and backend modernization, with experience working across SQL/NoSQL data stores, enterprise data platforms and event-driven systems. Comfortable owning backend architecture while remaining hands-on with implementation, testing, deployment, troubleshooting and production support.

CORE SOFTWARE ENGINEERING

Python & Backend Engineering — **Python 3.x, FastAPI, Django**, asynchronous/backend processing**, Node.js, Next.js**, REST APIs, GraphQL, service-layer design, validation, structured data processing, caching, rate limiting and production error handling.

GEN AI/Agentic Systems — Multi-agent orchestration, LangGraph state machines, LangChain, tool/function calling, MCP tool integration, structured output contracts (Pydantic / JSON schema), prompt engineering, self-repair and bounded retry loops, human-in-the-loop approval gates, Azure OpenAI, Amazon Bedrock, Claude API

Rag & Knowledge engineering — Hybrid retrieval (vector + BM25), semantic chunking, embeddings and vector stores, re-ranking, context assembly with source citations, grounded generation, hallucination detection, PII and policy guardrails, evaluation harnesses and human feedback loops, Graphana and LangSmith observability.

Workflow Orchestration — Production workflow design using Azure Durable Functions and AWS Step Functions, including parallel execution, state management, retries, idempotent activities, scheduled workloads, failure recovery and long-running workflow coordination.

Microservices & Distributed Systems — Designed and maintained 12+ production microservices and integration services. Event-driven and serverless patterns, service boundaries, API contracts, retries, idempotency, graceful degradation and third-party system integration.

Cloud Engineering — GCP: Cloud Functions, Firebase, Firestore and serverless workloads. AWS: Lambda, Step Functions and event-driven architectures. Azure: Functions, Durable Functions, Azure SQL, Blob Storage, Databricks, Data Factory and Key Vault. Docker and automated DEV/UAT/PROD deployment.

Data & Integration Engineering — Enterprise integrations across SQL/NoSQL databases, data platforms and third-party APIs including Power BI, Databricks, Microsoft Fabric, Xero, MYOB and Workday Adaptive Planning. Batch, scheduled and event-driven processing patterns.

Testing, DevOps & Reliability — Docker based deployment, Git-based delivery, CI/CD, environment promotion, automated deployment, production observability, Grafana, tracing, alerting, logging and incident support.

Security — OAuth/OIDC-aligned identity patterns, Entra ID, RBAC, service principals, secrets management, access-controlled API/data processing and least-privilege system design.

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer - SaaS Platform Architecture — Findex Australia

Melbourne · March 2022 – Present

- Architected and delivered production backend services and workflow-driven enterprise SaaS platforms using Python, Django, FastAPI, GraphQL, REST APIs and cloud-native services.
- Designed distributed processing workflows using Azure Durable Functions with parallel execution, retry policies, state tracking and idempotent activities for high-volume regulatory document processing.
- Built Django and FastAPI backend services with authentication, authorization, caching, rate limiting, structured query execution, bounded failure recovery, tracing and production alerting.

- Designed Python processing pipelines for complex financial, payroll and regulatory workloads, combining structured extraction, deterministic business rules, validation and exception processing.
- Established API and integration patterns across Power BI, Databricks, Microsoft Fabric, SharePoint, Workday Adaptive Planning, Xero, MYOB and internal enterprise services.
- Designed scheduled and automated data-processing workflows for financial modelling, benchmarking, student analytics and board-report generation.
- Defined engineering standards across API integration, service boundaries, CI/CD, security, observability, environment promotion and production support.
- Led cross-functional engineering teams while remaining hands-on with Python, Django, FastAPI, GraphQL, cloud integration and backend architecture.
- Stepped in as interim DevOps lead during a critical production incident, stabilising backend services, improving CI/CD reliability, adding Grafana observability and reducing manual operational overhead.

Senior Software Engineer, Product Innovation, Backend Cloud Engineering— Quest Payment Systems

Melbourne · October 2017 – March 2022

- Led backend and cloud engineering from concept through production for Airpay App, a regulated POS App and Donation Point Go, a regulated contactless donation SaaS platform.
- Architected and maintained 12+ production microservices and integration services using Node.js, AWS Lambda and AWS Step Functions.
- Designed event-driven, auto-scaling serverless architectures across AWS and GCP to handle unpredictable payment and transaction workloads.
- Built third-party integrations with Xero, MYOB and payment ecosystem services using resilient API integration patterns.
- Developed backend APIs supporting web, mobile, POS and kiosk applications.
- Implemented Google Firebase Functions, Firestore/real-time data services and NoSQL persistence for Airpay and Donation Point platforms.
- Led a team of three while remaining hands-on with backend development, cloud architecture, monitoring, logging and production support.

Technology Lead / Full Stack Developer — Apptarix Mobility

Bangalore · June 2013 – July 2017

- Filed a provisional patent application 201641021557 — a recommendation engine combining commercial and learnt rules, with a method to measure which of the two is earning its place.
- Built an early NLP sentiment engine using cosine similarity in R and Python; integrated clustering and sentiment analytics into OTT, analytics and EdTech product lines. Managed a team of 5. Integrated R and Python analytics for clustering and sentiment analysis, converting raw user data into product intelligence and customer insight.
- Designed scalable data ingestion, scraping and processing services that expanded the product data ecosystem and supported analytics-driven features.
- Managed a team of 5 in a startup environment using Agile and Scrum; led architecture across analytics dashboards, OTT SaaS platforms and client-facing product lines including Teletango, PrepSuite and MystiKit.

BACKEND, WORKFLOW & DATA PLATFORMS DELIVERED

Payroll AI — Python Workflow Automation & Payroll Assurance SaaS Platform

Findex · Solution Architect and Hands-on Engineer

Designed and built a Python-based multi-stage processing platform that converts Modern Award and Enterprise Agreement documents into executable payroll rules, validates those rules against payroll data and surfaces exceptions through Power BI.

- Designed a four-stage workflow covering document ingestion, structured rule extraction, client-specific calculation generation, payroll execution and exception remediation.
- Built the processing pipeline in custom Python, including retrieval and chunking strategies designed to preserve legal clause references across dense regulatory documents.
- Implemented traceability from every calculated payroll outcome back to its originating legal clause, supporting explainability, auditability and business validation.
- Designed controlled change workflows that determine the blast radius of a rule update across pay runs and clients before material changes are approved.
- Implemented automated exception classification and rerun coordination, reducing the need for developers to manually intervene in individual payroll corrections.
- Converted under- and over-payment detection from periodic audit activity into a continuous processing and exception-management workflow with human approval for flagged cases.
- Delivered as an eligible R&D activity under the Australian DISR R&D Tax Incentive program.

WealthAI — Distributed Document Processing & Workflow Orchestration

Findex · Solution Designer and Full Stack Development

Designed a serverless processing platform for concurrently processing large volumes of regulatory documents using stateful orchestration and independently executable workflow activities.

- Designed end-to-end workflow orchestration using Azure Durable Functions, including parallel execution, dependency management, retries and persistent workflow state.
- Implemented idempotent per-document processing activities so failed or retried workflows could execute safely without duplicating downstream operations.
- Built structured schema validation to ensure extracted regulatory information complied with defined application contracts before downstream processing.
- Separated probabilistic AI extraction from deterministic business-rule execution so regulatory outcomes remained repeatable and explainable. Integrated workflow outputs with JSON, Excel and database services.
- Designed graceful-degradation and failure-handling patterns so individual document failures did not compromise the complete processing workload.
- Implemented complete workflow state tracking to support operational monitoring, troubleshooting and reruns.

KRM Remuneration Platform — SaaS Platform - Secure Data Processing & Calculation Services

Findex · Platform Architect and Delivery Lead

Architected the backend, data and security foundations for an enterprise remuneration platform processing sensitive financial, incentive and workforce data.

- Designed processing pipelines for short-term incentive, referral, forecasting and audit workloads across enterprise financial datasets.
- Implemented Entra ID SSO, RBAC and row-level security across application services and Power BI Embedded.
- Designed backend authorization controls that determine permitted data access before sensitive information enters downstream processing workflows.
- Established governed bronze, silver and gold data layers using Databricks, Microsoft Fabric and OneLake.
- Built the underlying remuneration calculation engine covering performance weights, thresholds, ramps, payaway curves, gateways, caps, floors and employee segmentation.
- Replaced spreadsheet-based remuneration calculations with a deterministic, auditable processing engine.
- Integrated application, analytics and data-platform components across controlled DEV, UAT and production environments.
- Supported production onboarding of 443 users and 129 people leaders under the platform's role and permission model.

PerformX — FastAPI Backend & Governed Data Query SaaS Platform

Findex · Solution Architect and Hands-on Backend Engineer

Designed and delivered a production FastAPI service that converts natural-language requests into validated queries against governed enterprise data, enforcing authentication, authorization and data-access policies before execution.

- Built backend APIs using FastAPI to handle identity resolution, query generation, validation, execution and response orchestration.
- Implemented Entra ID authentication and permission enforcement across multiple stages of request processing to prevent unauthorized access to financial data.
- Designed bounded retry and self-recovery logic where failed generated queries are corrected using runtime errors, revalidated and terminated safely when recovery is unsuccessful.
- Ensured authoritative values are always produced by the governed Power BI semantic model rather than generated by the language model.
- Implemented rate limiting and caching of workspace, user and permission metadata to reduce backend latency and unnecessary platform calls. Added production tracing, monitoring, structured error handling and alerting.
- Exposed the backend through both a React APP browser interface and Microsoft Teams.

EDUCATION, CERTIFICATIONS & IP

- Harvard Business School Online — Certificate in Business Analytics, 2023
- BE, Electronics & Communication — Visvesvaraya Technological University, First Class Honours, 2008
- Certifications — Azure for Data Engineering (Coursera); Machine Learning & Big Data; Generative AI; React with TypeScript. PMI Melbourne member. PMP, Anthropic Claude Developer, Snowflake and Databricks pathways in progress.